

NYC Taxi Data Warehouse

PostgreSQL OLAP and Neo4j Graph Analysis

Course	DM2526 - Data Management 2025/2026
Dataset	NYC TLC Yellow Taxi Trip Records
Period	January-March 2024
Stack	PostgreSQL 16, Python, Neo4j 5

PostgreSQL is the authoritative dimensional warehouse. Neo4j is a derived aggregate layer used to inspect taxi corridors, pickup hubs, payment preferences, vendor trends, and peak-demand hours.

1. Objective and Dataset

The objective is to support analytical questions about demand, revenue, payment behaviour, trip duration, and geographic movement. The source dataset contains Yellow Taxi trip records published by NYC TLC.

Stage	Trip records
Raw TLC records	9,554,778
Cleaned and loaded facts	8,448,046
Removed by validation	1,106,732

Rows with missing timestamps or locations, out-of-period pickups, invalid fares or distances, invalid passenger counts, and implausible durations are removed. Missing tip amounts are retained as zero. The pipeline preserves 88.4% of the original records.

2. Warehouse Design

The fact-table grain is one cleaned taxi trip. The measures are fare, tip, total amount, trip distance, passenger count, and trip duration. The dimensions implement the proposed time, pickup, dropoff, payment, and service hierarchies.

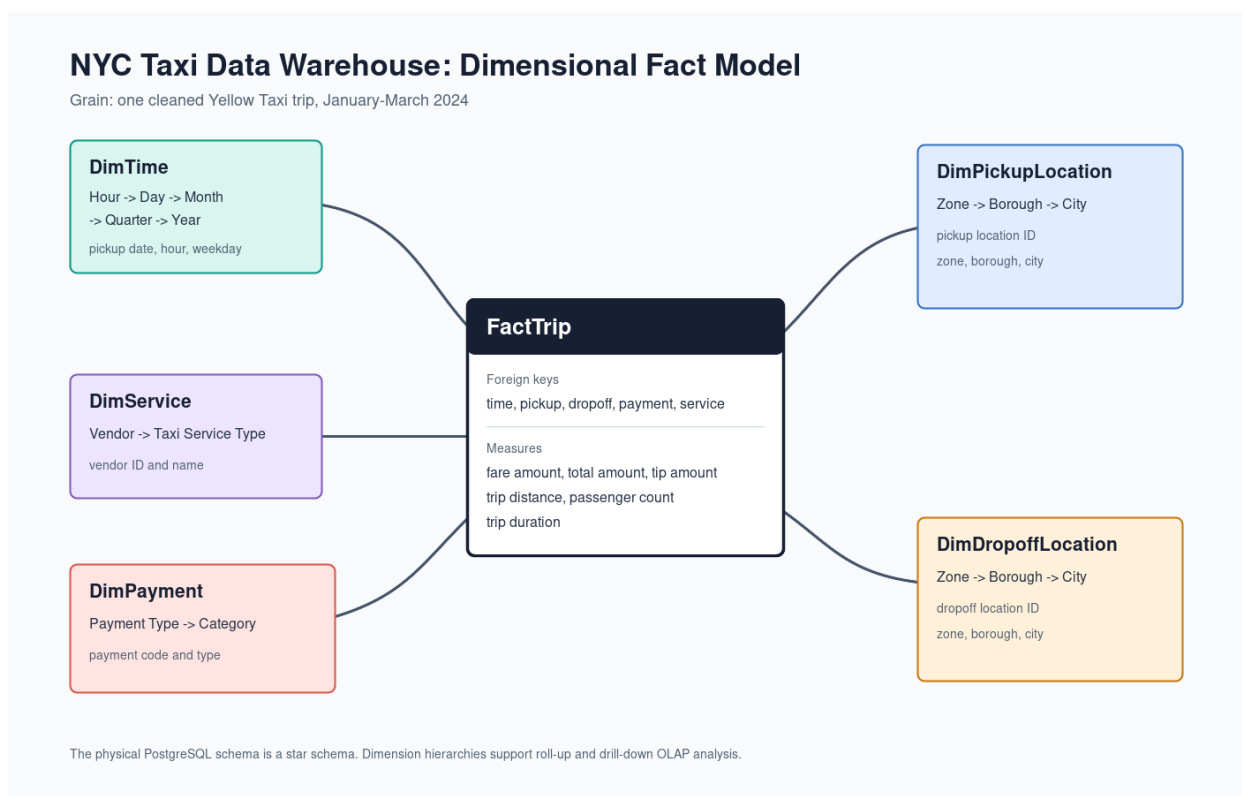
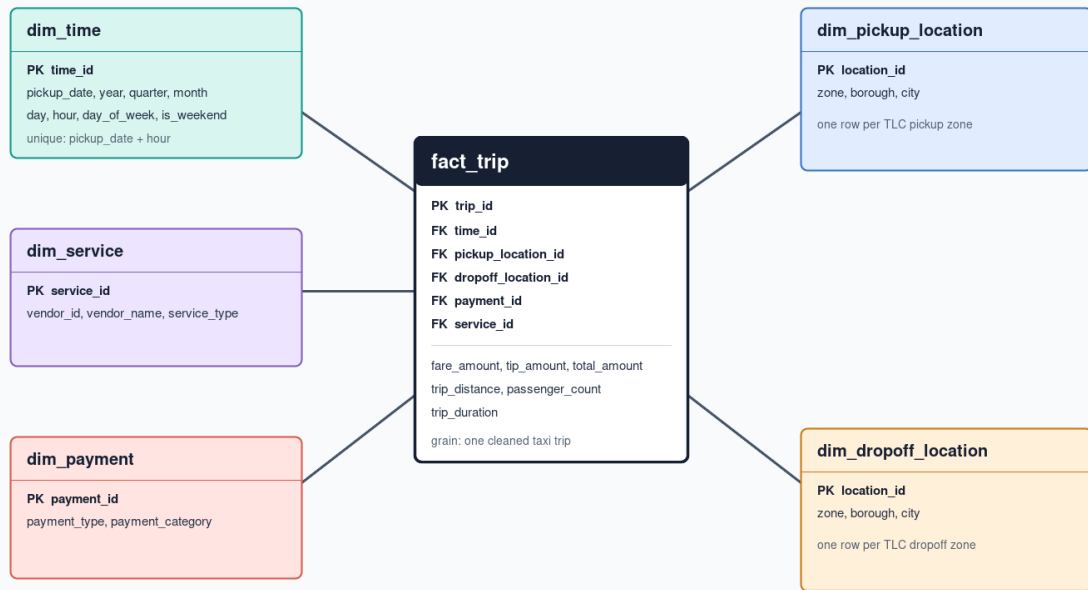


Figure 1. Logical Dimensional Fact Model.

NYC Taxi Data Warehouse: PostgreSQL Star Schema

Physical table names and foreign-key relationships



All dimension relationships are one-to-many: each fact row references exactly one row in each dimension.

Figure 2. Physical PostgreSQL star schema.

3. ETL and OLAP

The ETL pipeline downloads source files, cleans the records, and loads the five dimensions before bulk-loading fact rows in 100,000-row batches. The loaded warehouse contains 2,183 time rows, 265 pickup zones, 265 dropoff zones, six payment types, two services, and 8,448,046 facts.

Twelve reproducible SQL queries implement roll-up, drill-down, slice, dice, ranking, and window analyses. They examine revenue by time, demand by geography and hour, duration, payment preferences, tipping, and vendor performance.

4. Supporting OLAP Visualizations

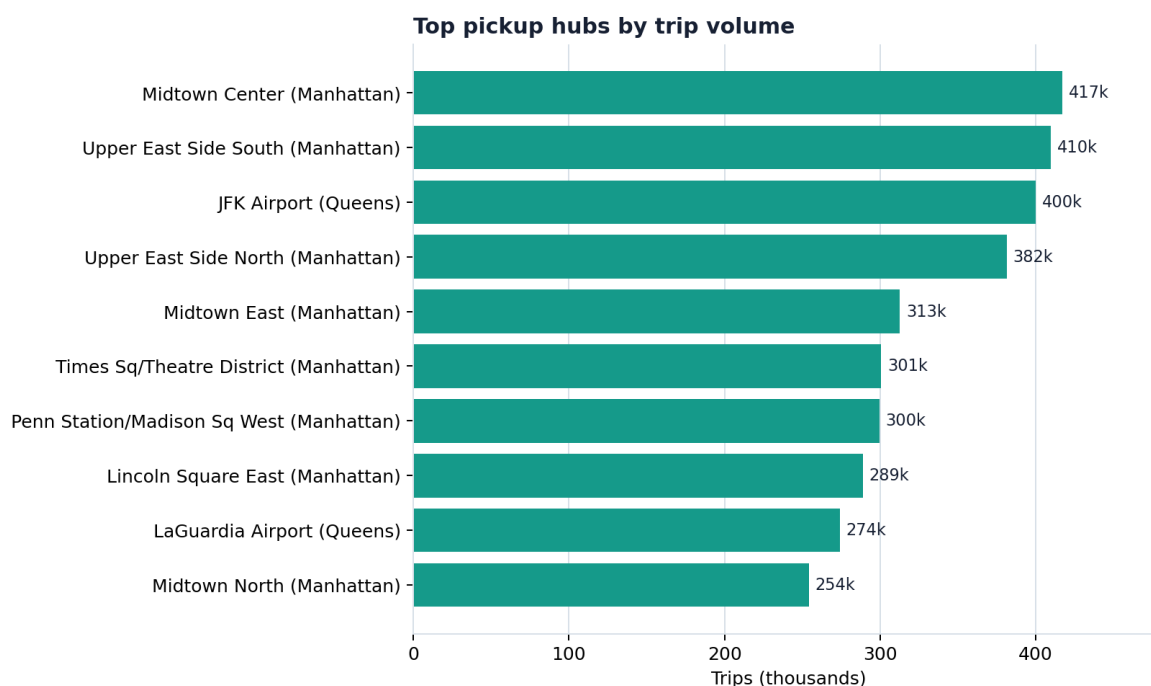


Figure 3. Top pickup hubs by trip volume.

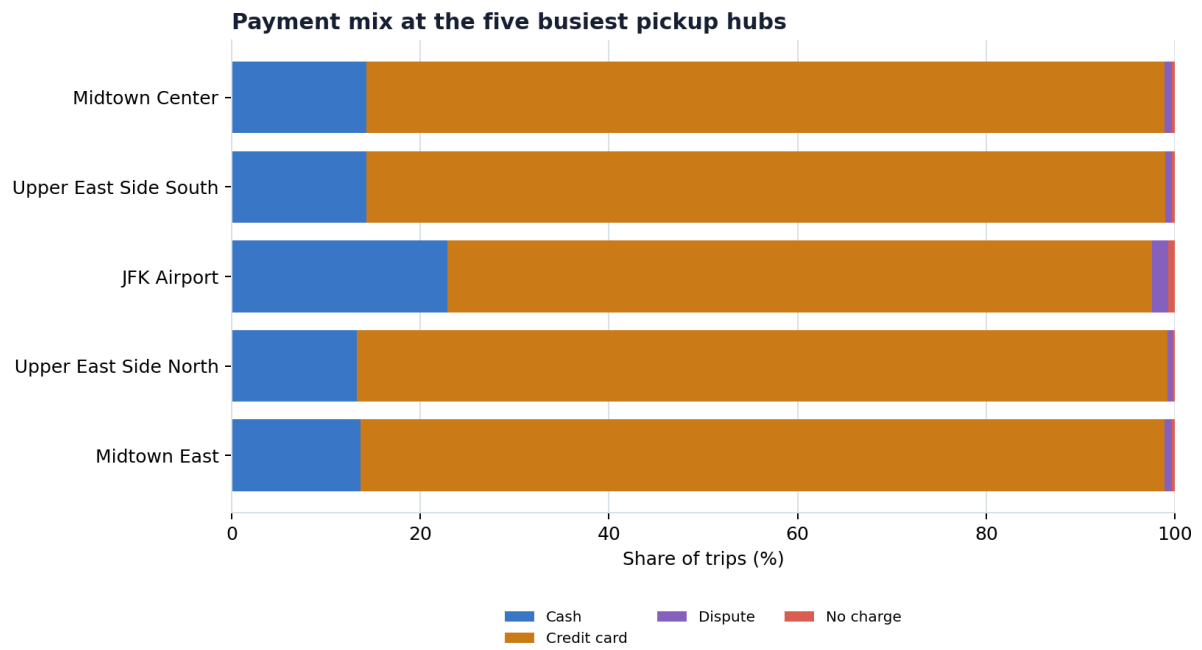


Figure 4. Payment mix at the five busiest pickup hubs.

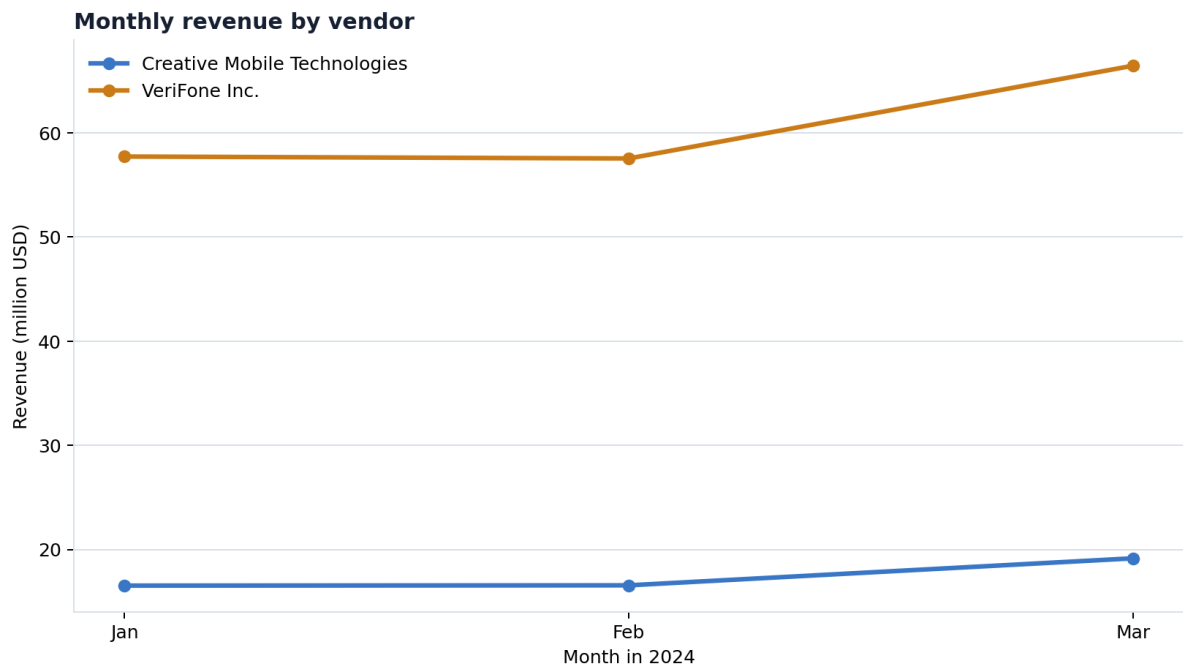


Figure 5. Monthly revenue by vendor.

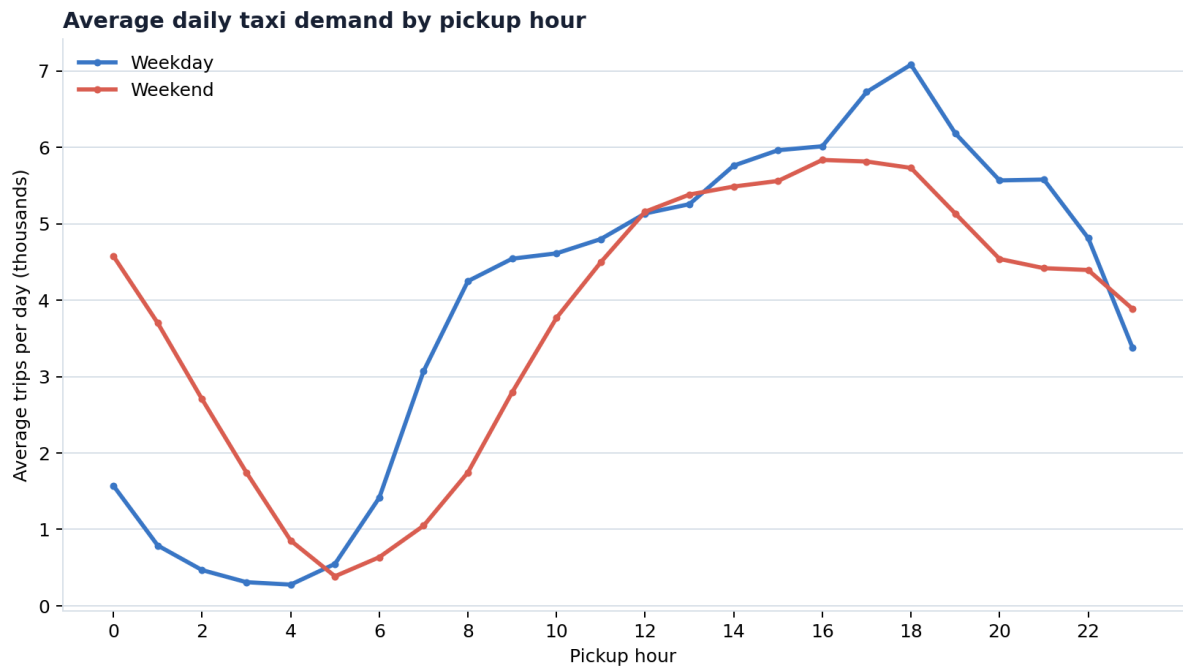


Figure 6. Average daily taxi demand by pickup hour.

5. Neo4j Graph Layer

Neo4j receives aggregates exported from PostgreSQL rather than individual trip nodes. The loaded graph contains 565 nodes and 39,590 relationships, including 33,304 directed TRIPS_TO relationships. This makes route structure and hub behaviour easy to inspect while retaining a reproducible source in the warehouse.

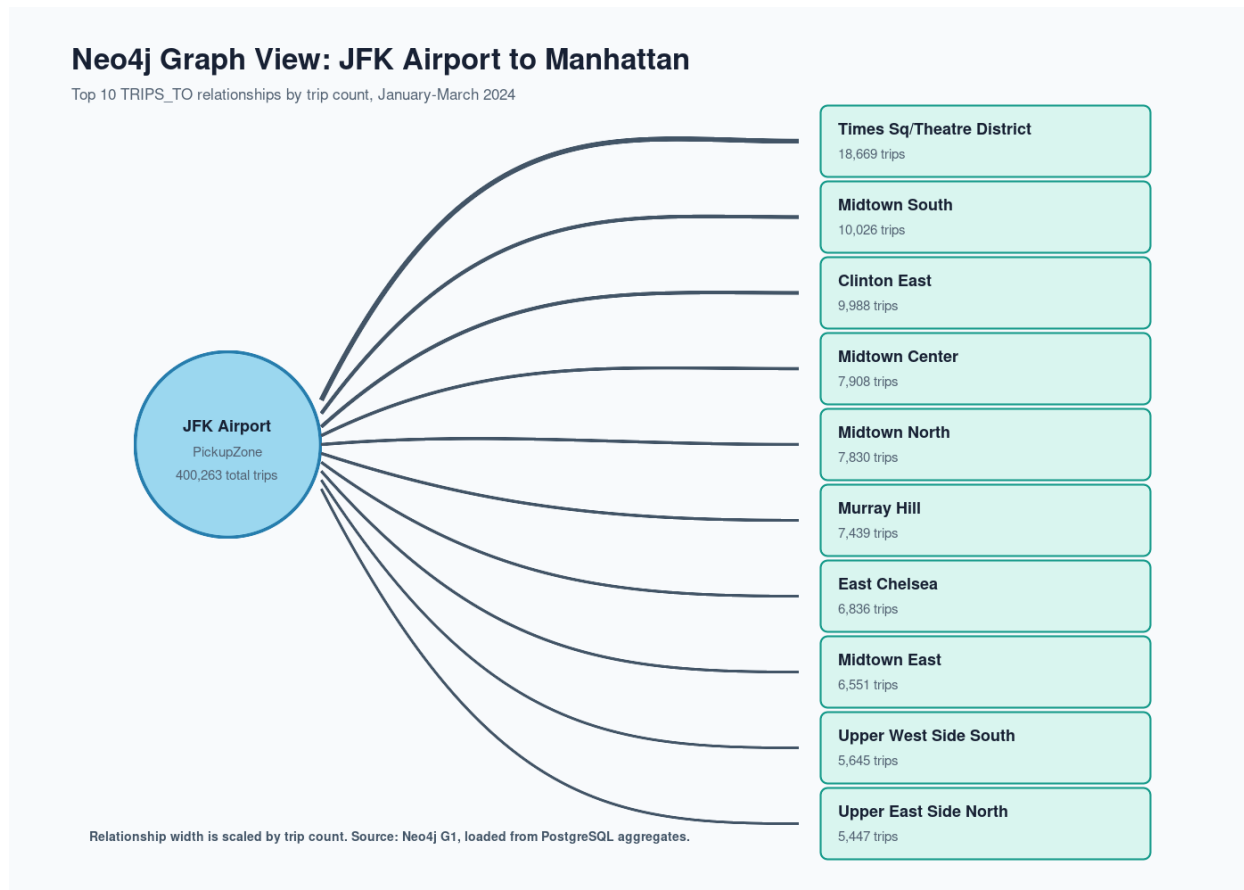


Figure 7. Top JFK Airport to Manhattan corridors from Neo4j G1.

6. Results and Interpretation

Finding	Evidence
Busiest corridor	Upper East Side South to Upper East Side North: 61,421 trips and \$958,770.89 revenue.
Largest pickup hub	Midtown Center: 417,379 trips. JFK Airport: 400,263 trips and \$32.88 million revenue.
Payment behaviour	Credit card is the preferred payment type at the busiest pickup zones.
Vendor trend	Both vendors record their highest revenue in March; VeriFone leads all three months.
Peak demand	Midtown Center peaks at 18:00 with 38,697 trips.

The star schema provides complete, auditable dimensional results. The graph layer adds an intuitive view of directional movement and network-like hub patterns. Together they satisfy the data-warehouse objective and make the analysis easy to demonstrate.

7. Reproducibility

Run the warehouse ETL, execute `sql/olap_queries.sql`, then run `etl/export_neo4j_graph.py` followed by `neo4j/load_graph.cypher`. The README contains the exact commands, while `docs/validation.md` and `docs/graph_analysis.md` record verified counts and query results.

8. Conclusion

The project implements the proposed warehouse workflow end to end: source data acquisition, cleaning, dimensional loading, OLAP analysis, data-model documentation, and visual relationship exploration. Neo4j extends the PostgreSQL warehouse without replacing it, giving the final submission both dimensional and graph perspectives on NYC taxi travel.